Presentation on Internship
completion at

K-AKA TECHNOLOGY SERVICES

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1. COMPANY PROFILE

1.1 Overview of the Organization

- **Company Name**: K-AKA Technology Services
- **Head of the Company:** Mr Abhishek Kumar Anshu
- **Established in**: 2020
- **Location**: 1st floor , Outer Ring road, Banaswadi, Bengaluru 560043, Karnataka
- **Industry**: Software Development & Cloud-based Services
- **Type**: Privately held SaaS company
- **Focus**: Software solutions and technical training

1.2 Vision and Mission

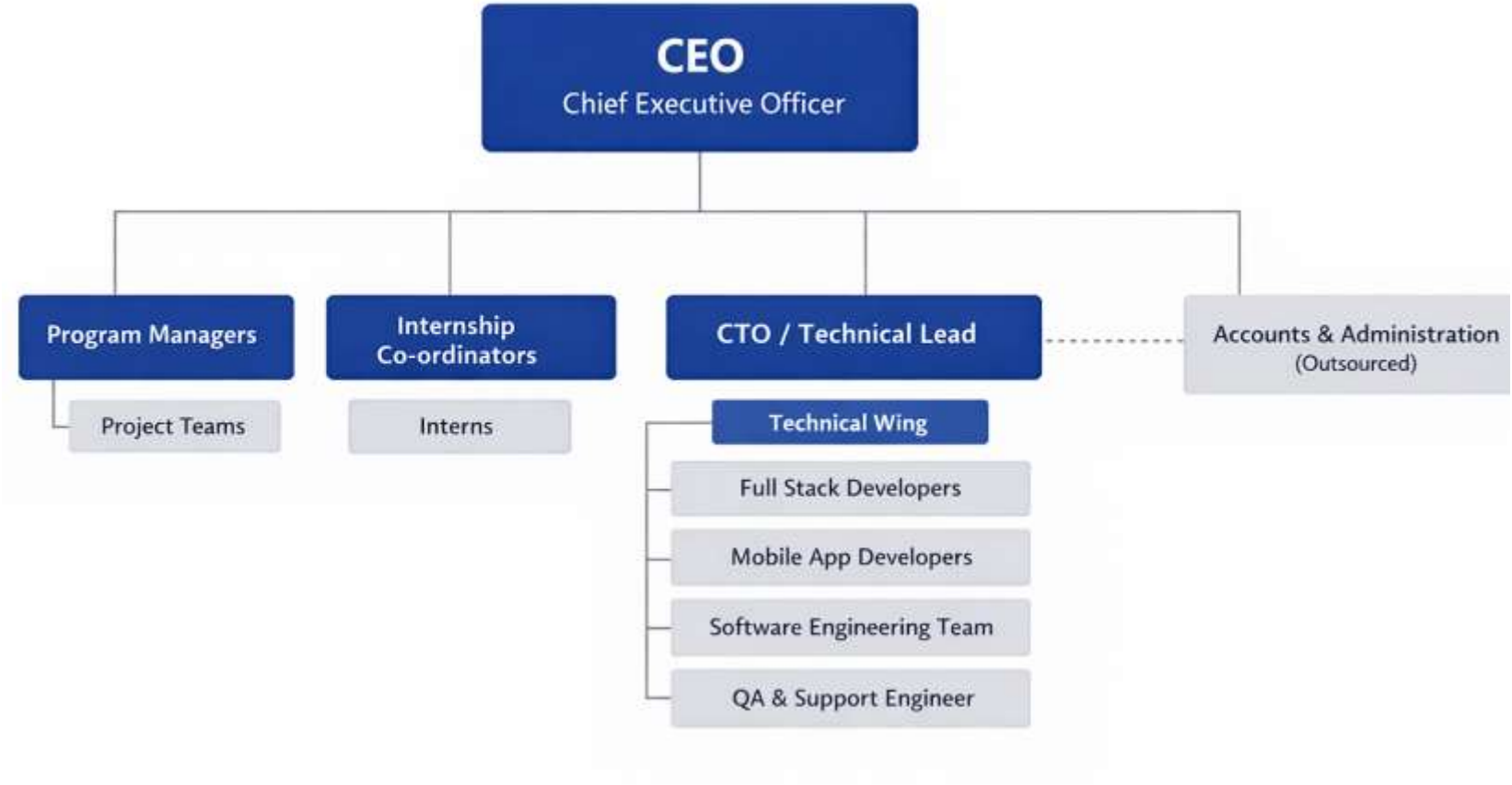
Vision

- “Our vision is to become the most trusted IT partner, creating smarter, more connected worlds by pushing the boundaries of technology.”

Mission

- “Our mission is to create customized, high-quality solutions that address our clients' unique challenges .”
- Deliver quality SaaS solutions
- Supporting businesses and learners through practical technology

1.3 Organization Structure



1.4 Roles and Responsibilities

- **CEO** - Strategic vision & decision-making
- **CTO** - Tech roadmap & architecture
- **Program Manager** - Project coordination
- **Software Engineers** - Core product development
- **Full Stack Developers** - Frontend & backend
- **QA & Support** - Testing & maintenance
- **Mobile App Developers** - iOS & Android apps
- **Sales & Business Dev** - Client acquisition
- **Accounts & Admin** - Finance & compliance
- **Internship Coordinators** - Intern guidance
- **Project Interns** - Development support

1.5 Products and Market Performance

❖ Products

- MyBusiness – Business Management Software
- KTS LMS
- iPC (Innovative Project Center)

❖ Services:

- Custom Web and Mobile Application Development
- System maintenance and support
- Technical training

❖ Client Portfolio

- rootmagics.com
- givingimpetus.com
- atomm.in
- skilledhire .in

❖ Market Performance

- Successful delivery of multiple web & mobile projects
- Serves startups, SMEs, and enterprises
- Growth in recurring maintenance contracts
- Expanding client base through referrals and digital presence
- ISO 9001:2015 certified | Reg. No: KR-03-0010654

K-AKA Technology Services is a growing SaaS company delivering reliable software solutions while fostering skill development through structured internship programs.

2. ON JOB TRAINING - 01

AIML Internship

2.1 Overview of Job Role

Job Role: AI/ML and Software Development Intern
Responsibility: Development
Objective: Build a machine learning-based web application for genetic disease risk prediction
Core focus areas: User authentication, dataset handling, backend integration, ML workflow

2.2 Skills Applied

Skill/Tool	How it was applied
Python	Wrote backend and frontend logic , API endpoints, data processing scripts and user interfaces
Pandas	Cleaned datasets — handled missing values, removed duplicates, encoded data
Scikit-learn	Implemented initial ML models for prediction
Flask	Designed REST APIs for auth, dataset upload, prediction requests
Firebase	Set up real-time database for user credentials and profile storage
Postman	Tested and debugged API endpoints between frontend and backend
HTML/CSS/JS	Built login, signup, and dashboard interfaces

2.3 Tasks Performed and Implementation

System Development:

- User authentication (login, signup, CAPTCHA)
- Firebase database integration
- UI for login/signup/dashboard
- Frontend-backend communication via HTTP/APIs
- Dataset upload module (Excel format)

ML & Data Work:

- Data pre-processing (missing values, duplicates, encoding)
- ML prediction interface
- Basic visualization (charts, risk indicator)
- API testing and debugging via Postman

2.4 Challenges , Solutions and Learning

Challenge	How it was solved	What was learnt
Inconsistent dataset formats	Data cleaning, normalization, encoding	Effective data pre-processing
Frontend-backend API errors	Postman testing, debugging request-response	Backend API development
Complex UI vs. functionality	Iterative UI refinements	User Centric Design

2.5 Learning Outcomes and Value Addition

Value Added to Organization:

- Functional prototype for genetic disease prediction
- Structured data pre-processing pipeline
- Improved data accessibility through simple interface
- Groundwork for a scalable predictive system

Learning Outcomes:

- ML model integration into web apps
- Designing web pages
- Debugging & problem-solving
- UI design for non-technical users

3. Use Case 01
Genetic Disease Prediction System

3.1 Title and Overview

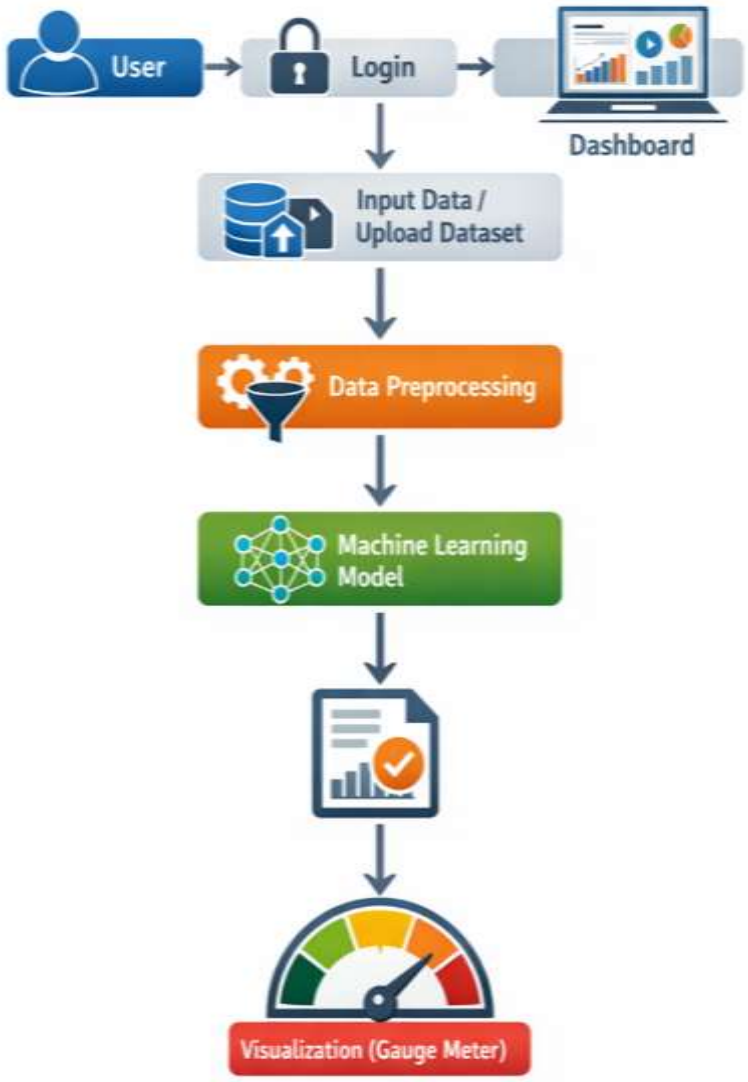
Title: Genetic Disease Risk Prediction System

Description: ML-based app to predict genetic disease probability using patient data and genetic indicators. Provides numerical + visual outputs.

Actors: User (Doctor / Researcher / Admin), System

Inputs: Age, Gender, Family History, SNP Values (SNP1–10), Gene Expression Values, Dataset (.xlsx)

3.2 System Workflow

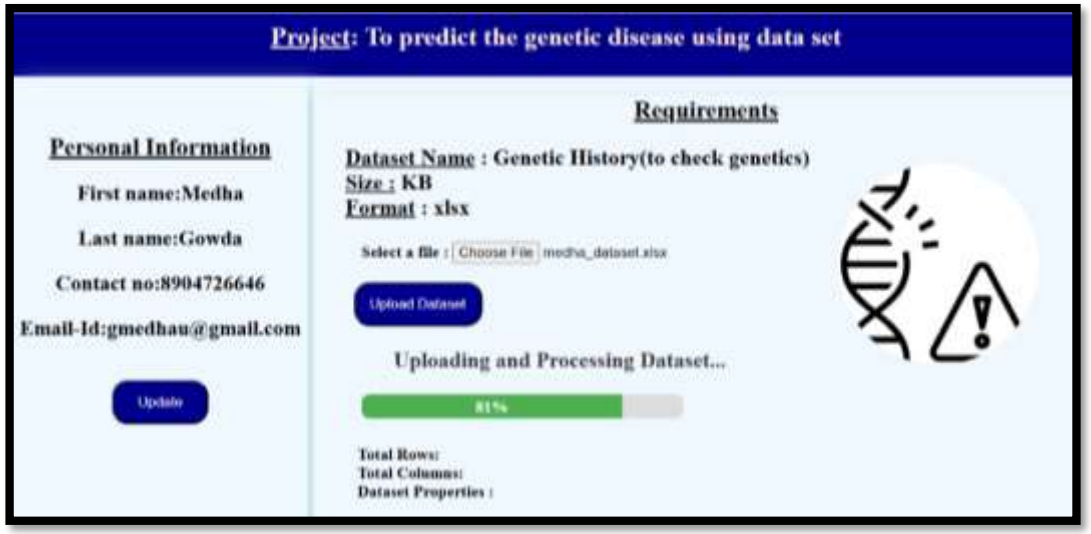


3.3 Implementation

Step 1: System Initialization

```
C:\Users\HOME\OneDrive\Desktop\fullstackmlproject\backend>py api.py
 * Serving Flask app 'api'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
```

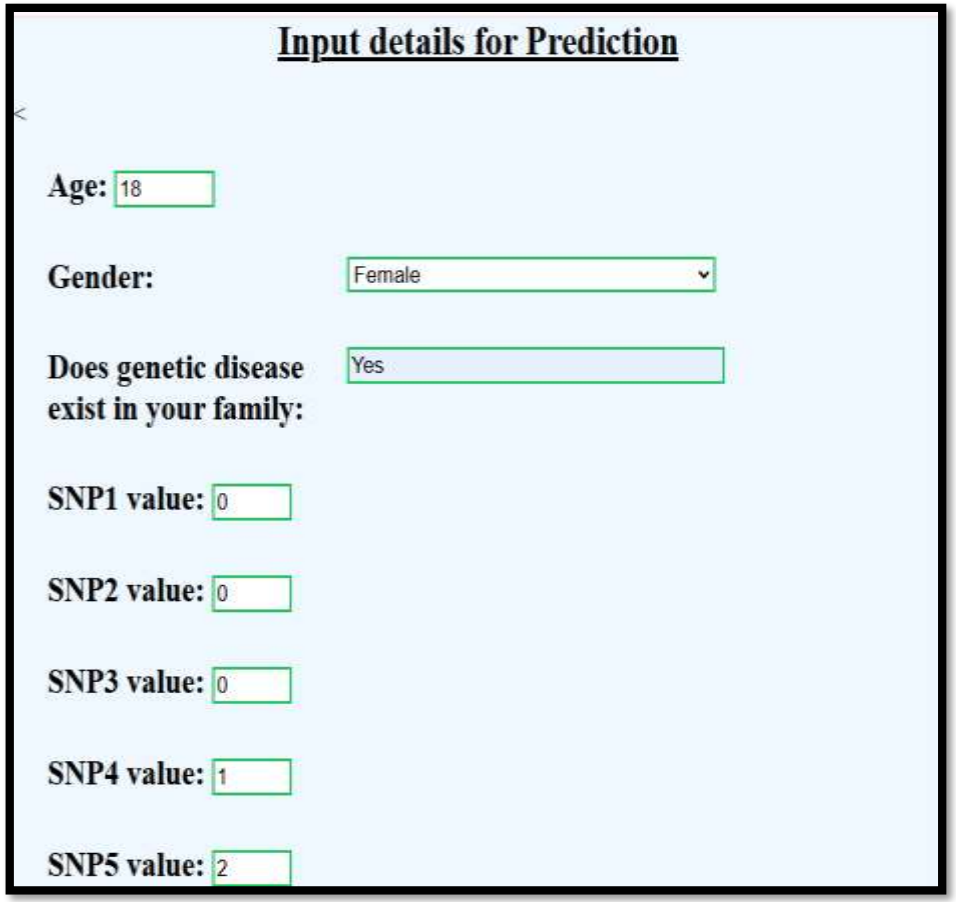
Step 2 : User Authentication



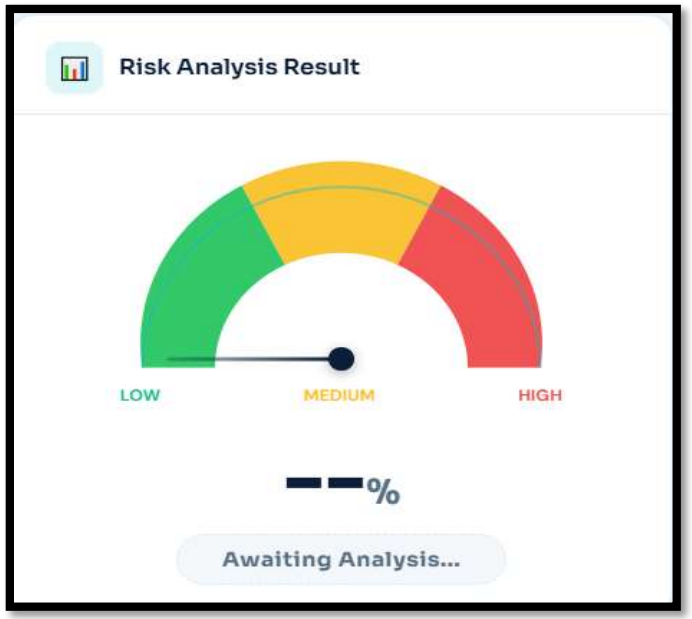
Step 3: Dashboard Access

-Dataset upload and preprocessing
-Calculation of Rows and columns
-Update button

Step 4: Prediction Inputs -> ML model processes data



Step 5: Output Visualization



Output generated: Risk %, Risk Category (Low/Medium/High), Graphical Visualization

3.4 Advantages, Limitations and Future Enhancements

Advantages	Limitations	Future Enhancements
Early genetic risk identification	Accuracy depends on dataset quality	Advanced ML models
User-friendly for non-technical users	Limited input parameters	OTP-based login
ML integration for prediction	Basic authentication only	Admin module
Dataset upload flexibility	Firebase security concern	Stricter database security

4. On Job Training 02

AIML Internship

4.1 Overview of Job Role

Role continues: AI/ML & Software Development Intern
Responsibility: Testing & Optimization
Goal: Enhance initial prototype into refined, user-ready application with improved accuracy and usability.

4.2 Advanced Tech & ML Models Used

Skill/Tool	How it was applied
XGBoost	Evaluated 5 models, selected XGBoost for best balance of accuracy & efficiency
Model comparison	Implemented bar chart visualization of model accuracies
Flask (enhanced)	Restructured APIs for better modularity and efficiency
Firebase (enhanced)	Improved CRUD operations, strengthened data security
Frontend (advanced)	Card-based responsive layouts, keyboard shortcuts, interactive elements
Feature engineering	Improved normalization, feature selection, dataset validation

4.3 Enhancement and New Features

New Features Added:

- Sample dataset download & format guidance
- Toggle-based SNP selection
- "Fill Sample Data" autofill
- Prediction history tracking
- Genetic information educational module
- Dynamic risk gauge meter
- PDF report generation

Enhanced Features:

- Improved authentication & validation
- Redesigned dashboard with navigation
- Profile update & account deletion
- Keyboard shortcuts (Ctrl+E, Ctrl+U, Ctrl+P)

4.4 Challenges, Solutions and Learning

Challenge Faced	How it was solved	What was learnt
Improving prediction accuracy over OJT 01	Transitioned from basic Scikit-learn models to XGBoost with improved normalization and scaling	Advanced ML algorithm implementation
Strengthening validation	Implemented strict validation checks — data stored only after successful verification	System testing, debugging & iterative improvement
Designing a user-friendly interface for complex inputs	Auto cursor pointing, toggle buttons, autofill, displaying keyboard shortcuts	User-centric interface design for non-technical users
Handling profile update without validation	Extended validation to all profile fields — email format, phone number, user ID	Security & input validation mechanisms

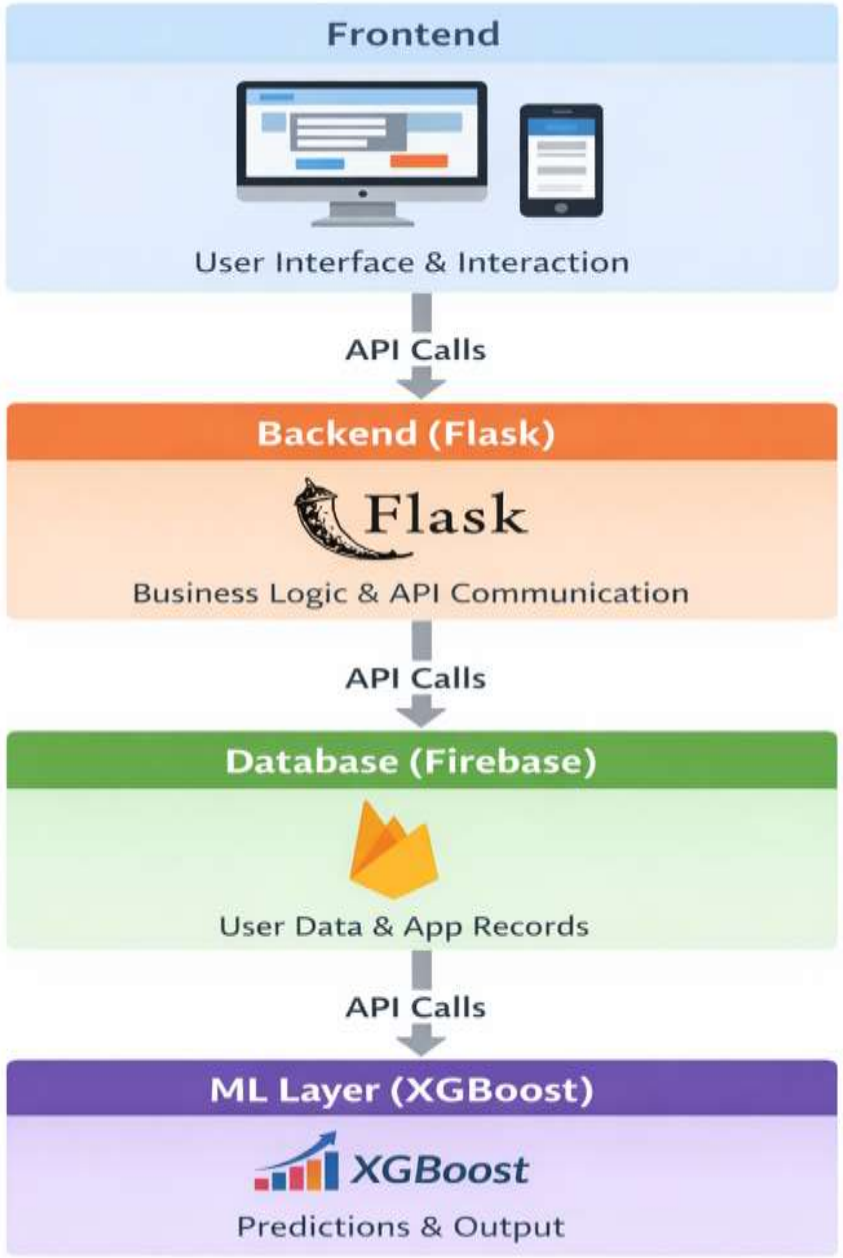
4.5 Learning Outcomes and Value Addition

Value Added to Organization:

- Improved model accuracy and reliability
- Intuitive UI with accessibility features
- Reduced dependency on external datasets (sample dataset)
- Prediction history + PDF reports for usability

Learning Outcomes:

- Model comparison & selection techniques
- Advanced ML (XGBoost) implementation
- User-centric interface design
- Security & validation strengthening
- End-to-end application development



5. Use Case 02
Enhanced Genetic Disease Risk Prediction System

5.1 Title & Overview

Title: Enhanced Genetic Disease Risk Prediction System

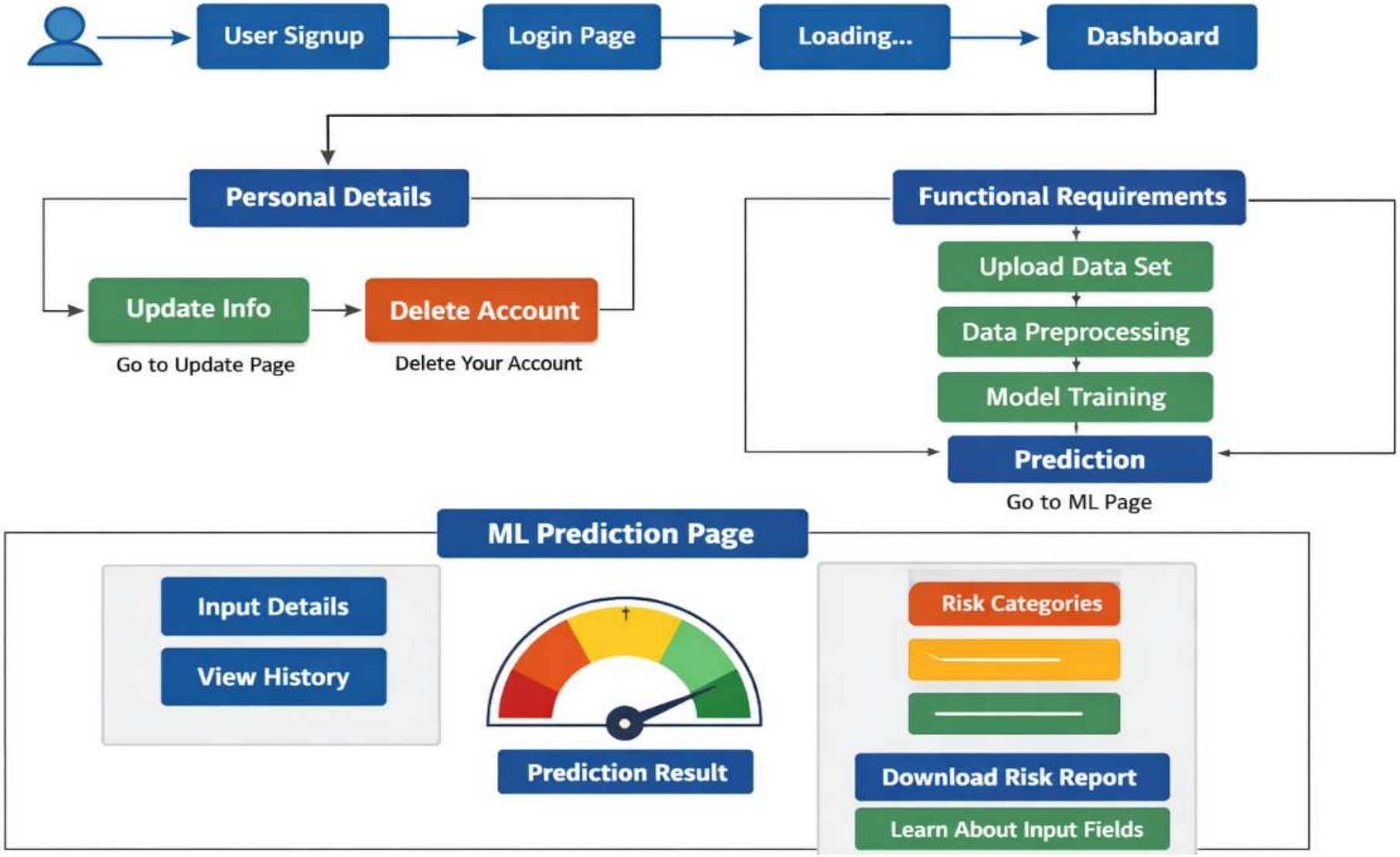
Description: Advanced version — improved accuracy, usability, new modules (history, dataset guidance, report generation)

Actors: User (Doctor / Researcher / Admin), System

Inputs: Same as UC01 + toggle-based SNPs, optional name field for history

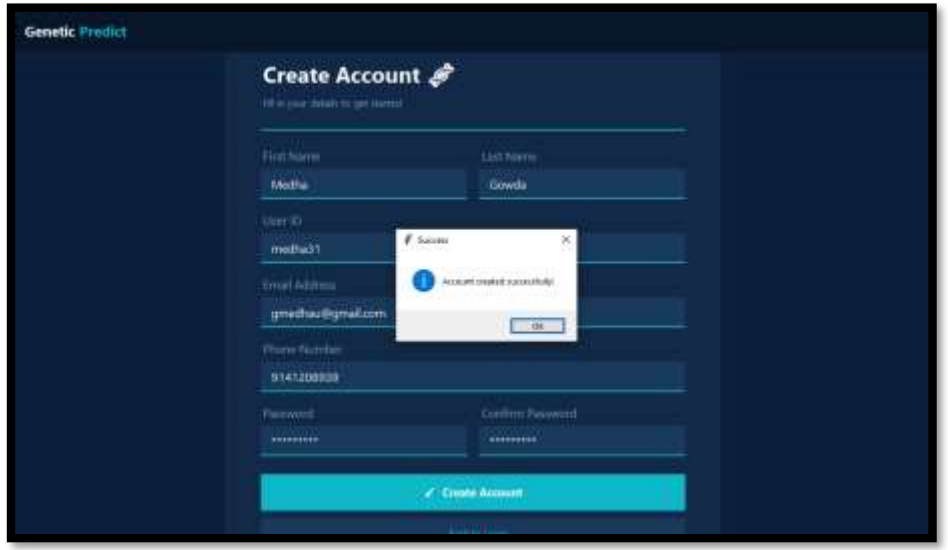
Key improvement over UC01: Full ML integration with XGBoost, complete end-to-end workflow

5.2 System Workflow

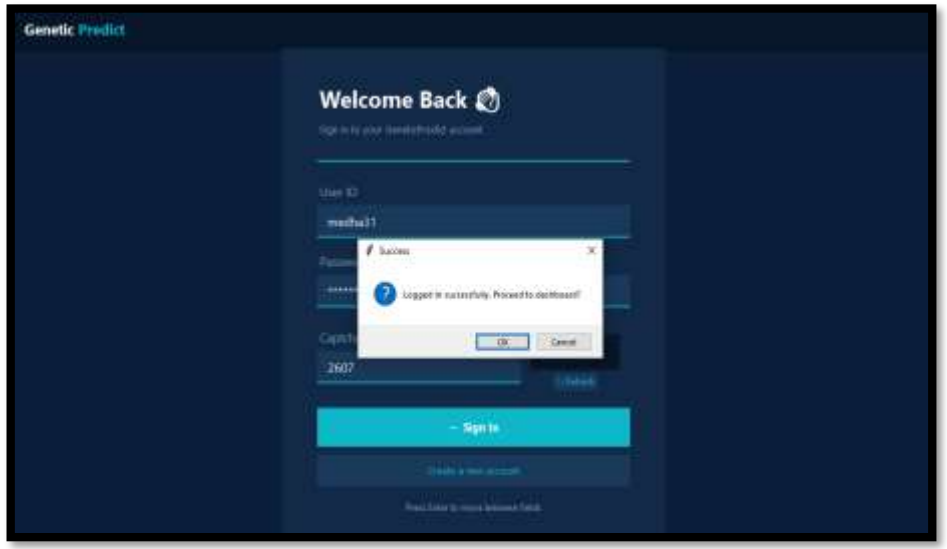


5.3 Implementation

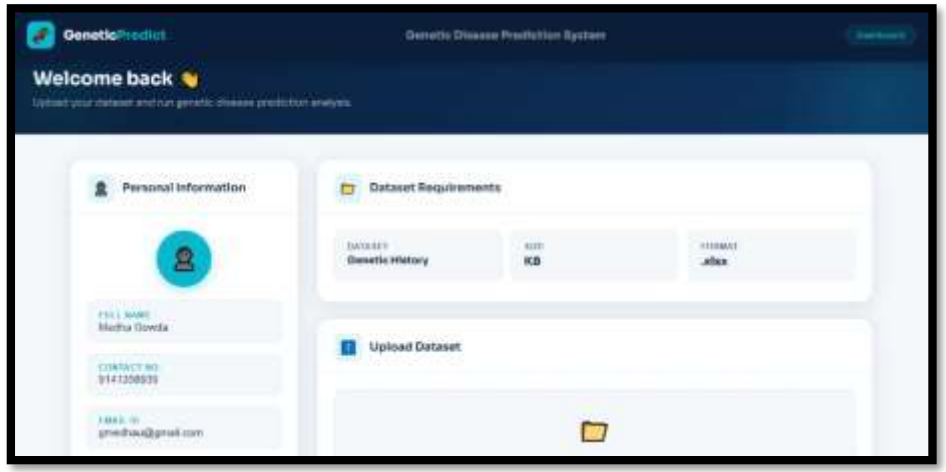
Signup Page



Login Page

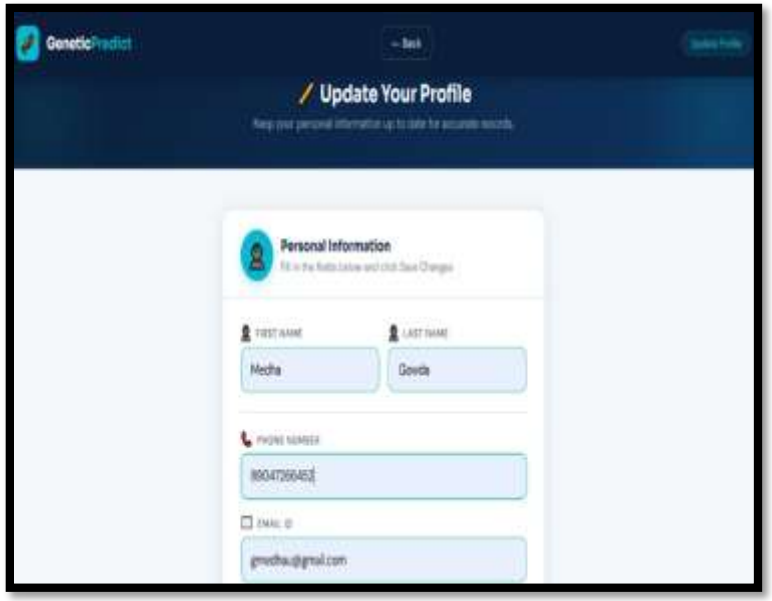


Dashboard

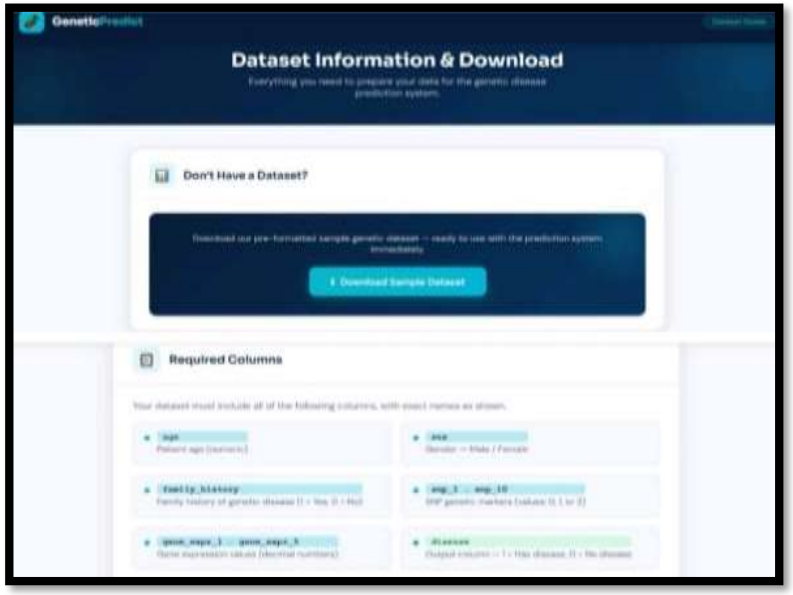


Personal Information

Update Information



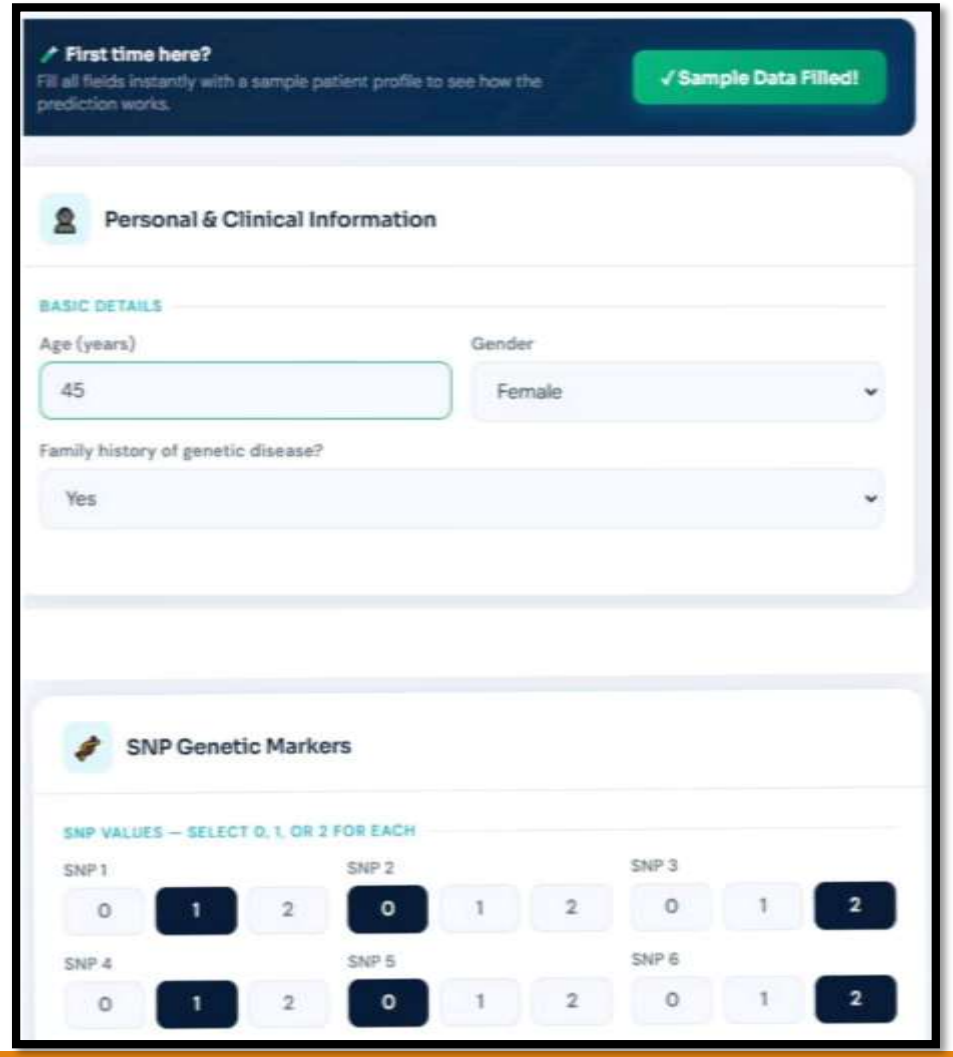
Dataset Info



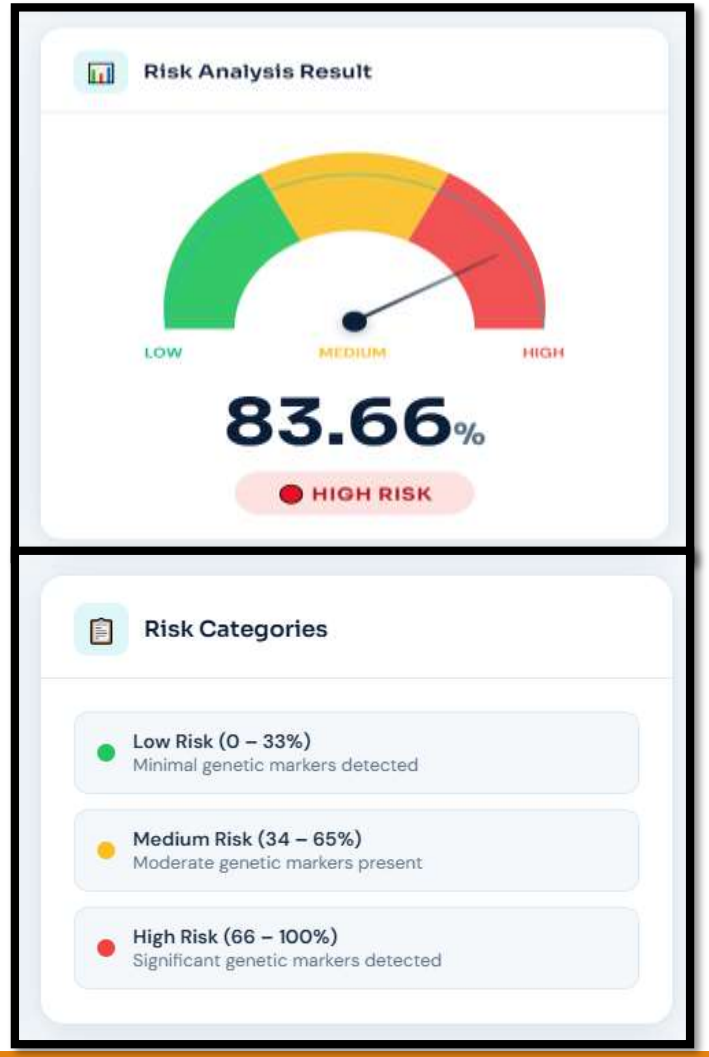
After dataset Upload

Prediction Module

Input fields



Final Output

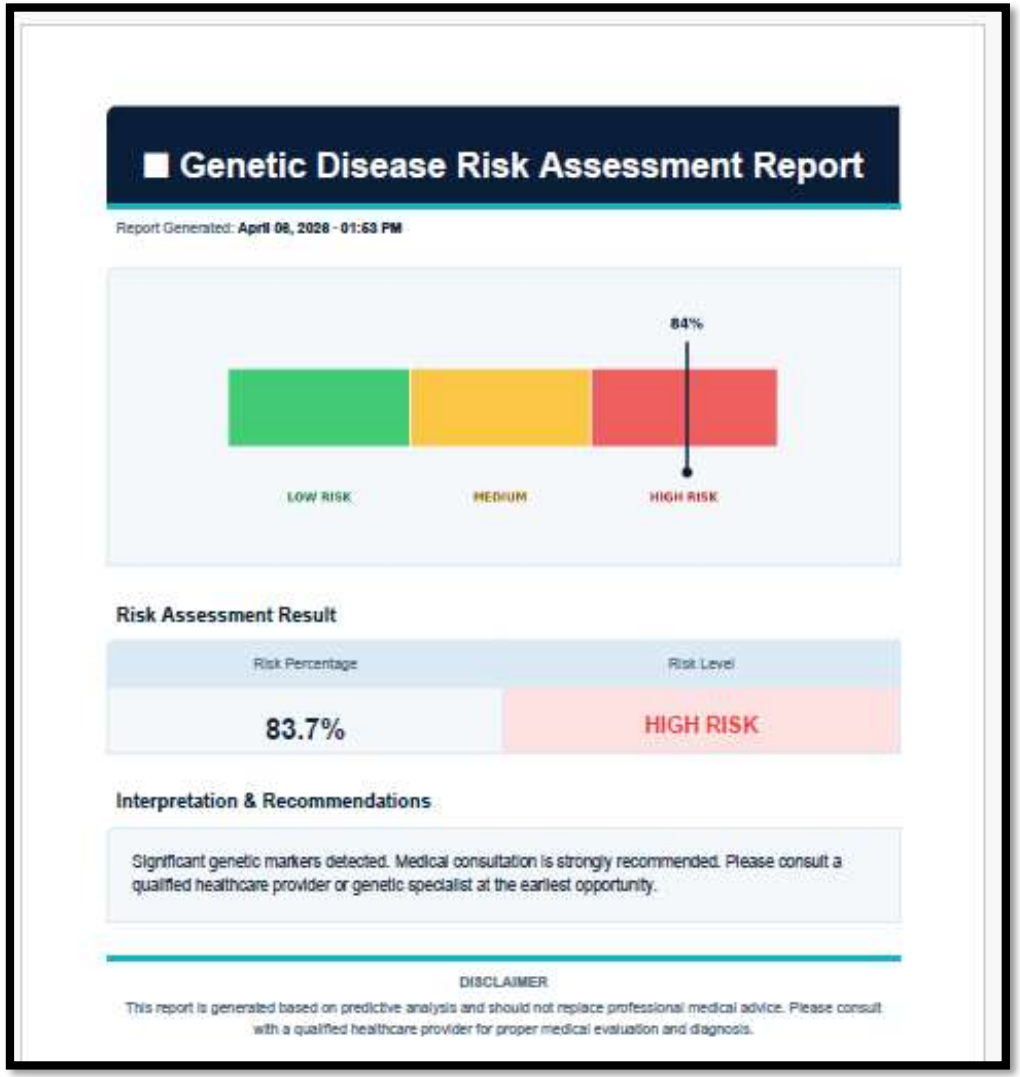


Added Features

Genetic Info

History Page

PDF Report



Genetic Disease Risk Assessment Report

Report Generated: April 08, 2026 - 01:53 PM

Risk Assessment Result

Risk Percentage	Risk Level
83.7%	HIGH RISK

Interpretation & Recommendations

Significant genetic markers detected. Medical consultation is strongly recommended. Please consult a qualified healthcare provider or genetic specialist at the earliest opportunity.

DISCLAIMER

This report is generated based on predictive analysis and should not replace professional medical advice. Please consult with a qualified healthcare provider for proper medical evaluation and diagnosis.

5.4 Advantages, Limitations and Future Enhancements

Advantages	Limitations	Future Enhancements
Complete end-to-end system	Dataset quality affects accuracy	Deep learning models
XGBoost for improved accuracy	Not a medical diagnosis substitute	Mobile app development
Strong validation & authentication	Complex biology not fully captured	Real-time healthcare integration
Prediction history & PDF reports	Full deployment not implemented	Cloud deployment (Render/Railway)

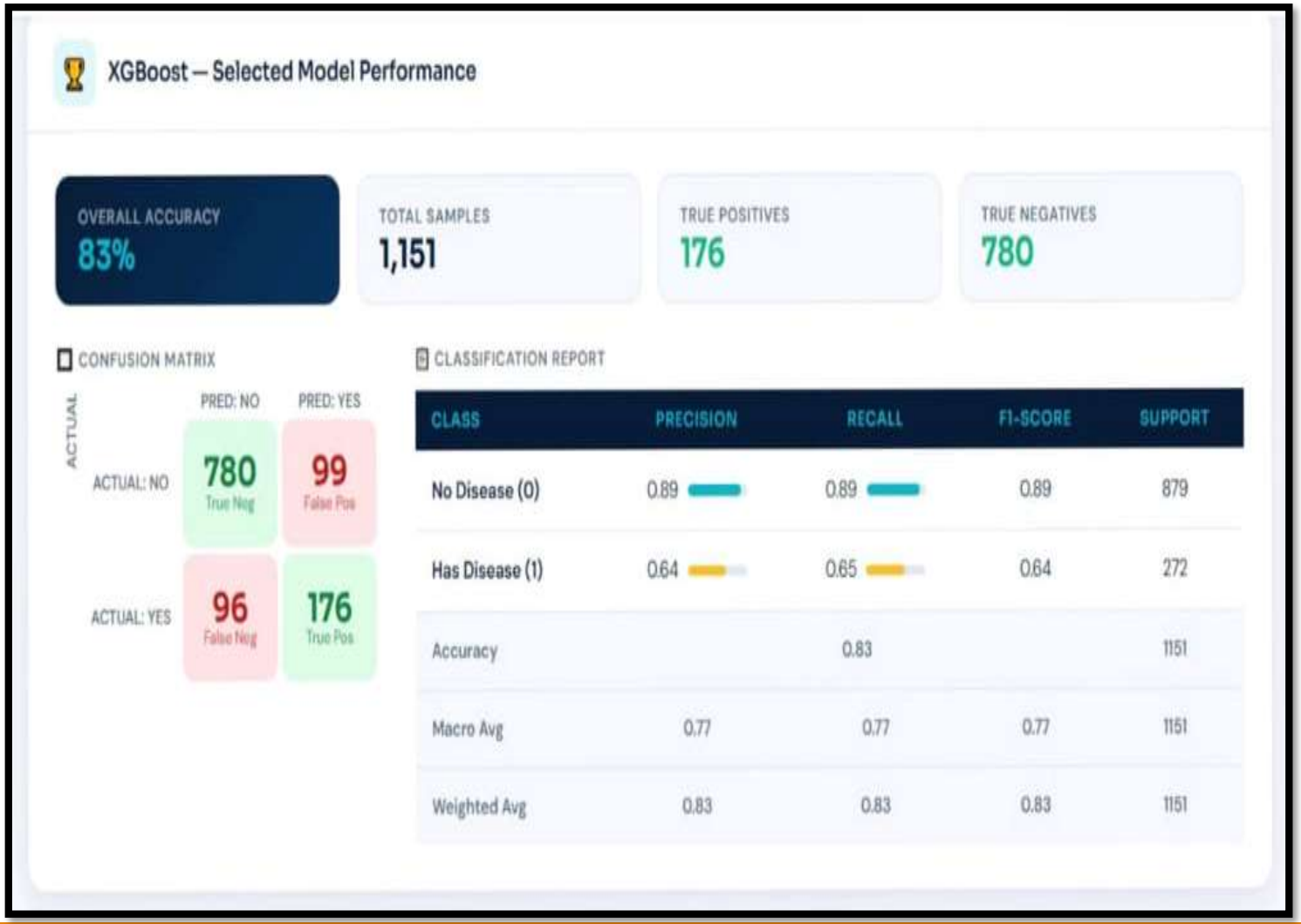
6. Model Selection and Evaluation

Multiple ML models were evaluated to find the best fit for genetic disease prediction
Evaluation based on: Accuracy, reliability, ability to handle non-linear data, scalability

Model	Accuracy	Reason Not Selected
Logistic Regression	80%	Cannot handle non-linear relationships
SVM	84%	Inconsistent probability outputs
Random Forest	82%	Moderate performance for this dataset
Gradient Boosting	83%	Less efficient than XGBoost
XGBoost	**83%**	**✓ Selected — Best balance of accuracy & efficiency**

6.1 XGBoost Performance Metrics

- XGBoost demonstrated consistent, reliable performance
- Slightly lower precision for Risk class is a known challenge with imbalanced datasets
- Can be improved with better dataset balance and feature representation in future



7. Conclusion

This internship at **K-AKA Technology Services** provided hands-on experience in software development, ML, and system design across two phases.
The **first phase** built foundational skills in authentication, APIs, and ML interfaces, while the **second phase** delivered a production-ready application with XGBoost, enhanced UI, prediction history, and PDF reports. **Use cases** demonstrated the evolution from basic static predictions to a secure, end-to-end workflow with live visualization and reporting.

Overall, the experience strengthened full-stack development, ML, and system design skills, offering valuable industry exposure and practical problem-solving experience.